



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID BP Physics

Magnets and magnetic materials

Class Outline

Title of Unit	Introduction to magnetic field
Title of Lesson	Magnets and magnetic materials
Starter Activity	Demonstration: magnetic field near a permanent magnet
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- demonstration: magnetic field and direction of field - Inquiry: compare and contrast a permanent magnet and an electro magnet
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- work on the worksheet as pairs - Use AI properly to find out use of permanent and electromagnet
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: • Understand magnets and magnetic material (Iron, Nickle and cobalt) • Draw and interpret magnetic field strength and field lines • Differentiate soft and hard magnetic materials • Understand ow an electromagnet works
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Thinking: Applying key ideas and facts in new contexts – apply understanding of E field in B field
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer, thinker
Teaching Activities	- demonstration: visualize magnetic field lines - worksheet: plot magnetic field lines (with help from a simulator); compare and contrast permanent and electromagnets - Worksheet: vocab check
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Check for understanding: key concepts, right hand and left hand rule

Vocabulary



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- Magnet 磁铁
- Magnetic material 磁性材料
- Alloy 合金
- Attract 吸引
- Repel 排斥
- Magnetic poles 磁极
- Magnetic field 磁场
- Compass 指南针
- Iron 铁
- Nickle 镍
- Cobalt 钴
- Induced magnetism 感应磁性
- Electromagnet 电磁铁



Magnets and magnetic materials

- Magnets can exert a force on another magnet or on magnetic materials.
- While magnetic materials are not magnets, they are attracted to magnets.
- There will always be a force of attraction between a permanent magnet and a magnetic material

You need to remember the names of these magnetic materials:

- iron
 - nickel
 - cobalt.
-
- Two magnets may either attract or repel or each other depending on the poles
 - Many materials are non-magnetic, meaning that they do not experience a force when in a magnetic field. Could you name a few non-magnetic materials?

Magnetic poles



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- Every magnet has a north and a south pole, positioned at opposite ends of the magnet.
- If you cut a magnet in half, you get two smaller magnets, each with a north (N) and a south (S) pole
- Opposites poles attract and the similar poles repel

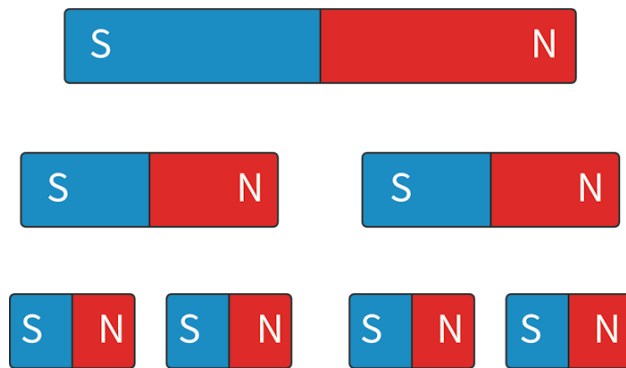


Figure 1. A magnet being cut in half twice.

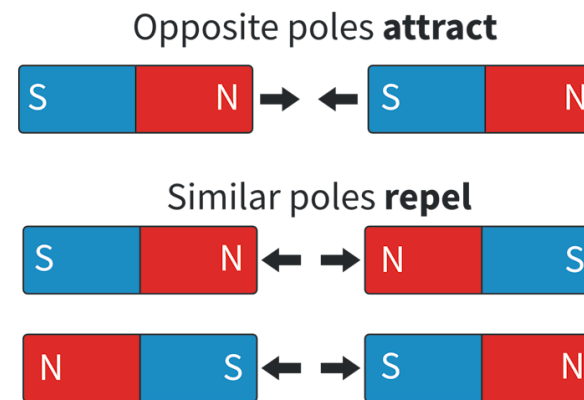


Figure 2. Attraction and repulsion of poles with force arrows.

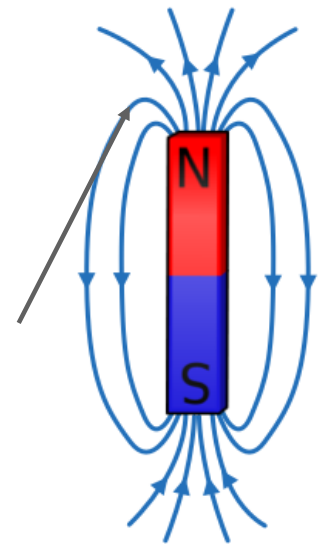
What is a Magnetic Field?



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- Every magnet is surrounded by a magnetic field
- A magnetic field is a region of space where another magnet or magnetic material experiences a force
- The magnetic field is a vector.
- The direction is given by the tangent to the curved magnetic field at any point.
- By convention, the direction of the magnetic field at any point is the direction that the north pole of a compass points towards.

The density of lines indicates how strong the field is.



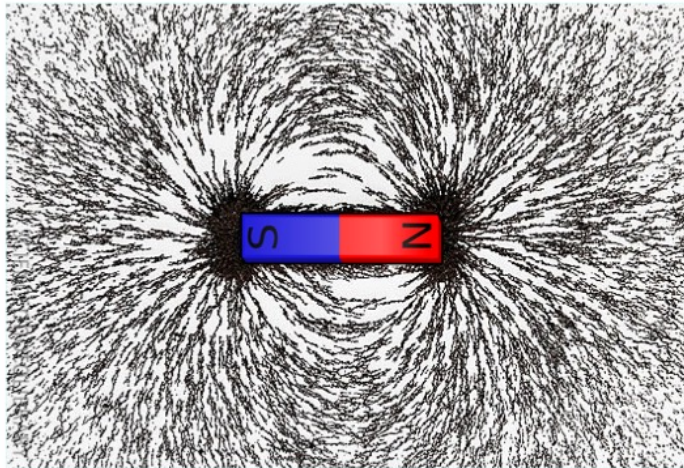
Detecting Magnetic Field Lines



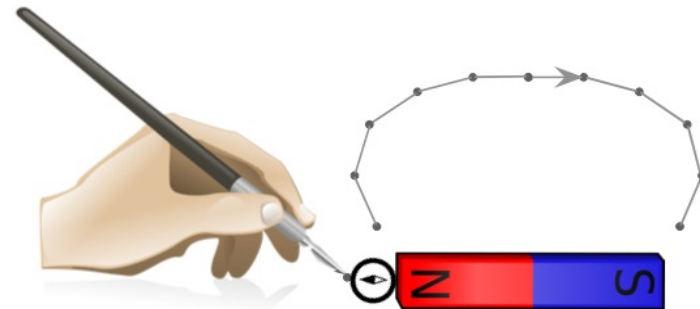
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Magnetic field lines are invisible. You can detect magnetic field lines using:

Iron Filings



Plotting Compass

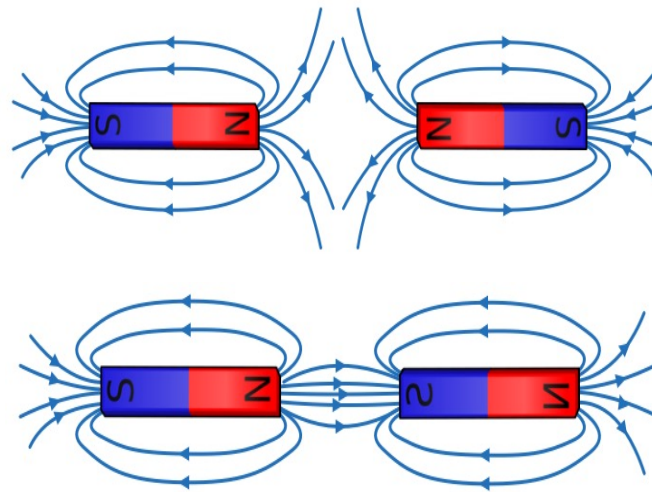
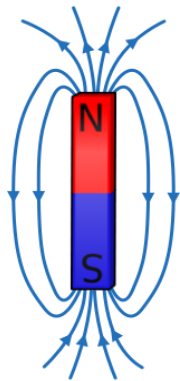


Magnetic Field Patterns



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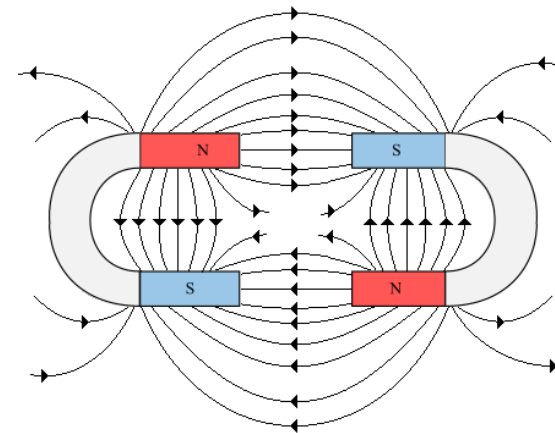
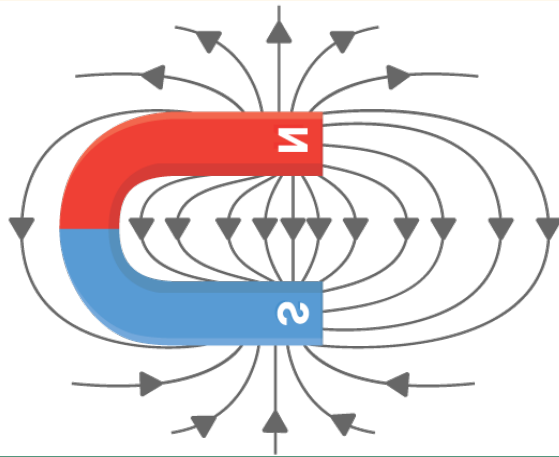
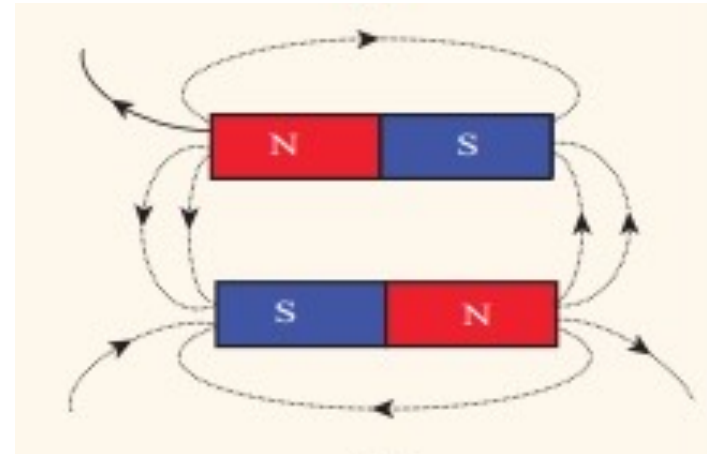
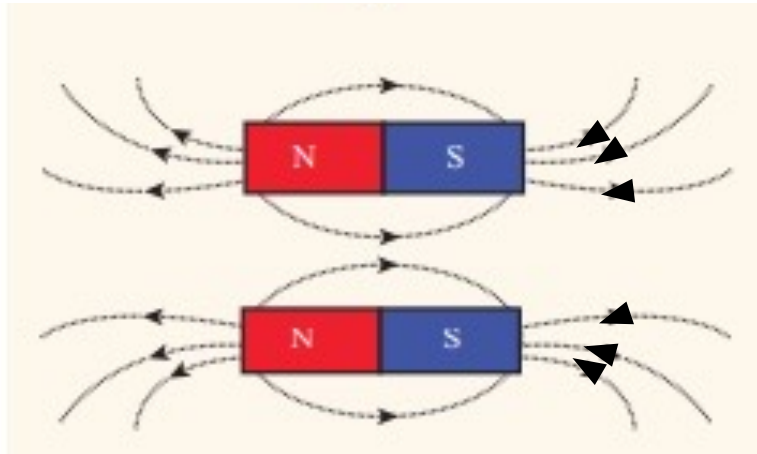
- Different configurations of bar magnets will produce different patterns of magnetic fields
- Magnetic field lines always exit from a north (N) pole and enter at a south (S) pole



Magnetic Field Patterns - more



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Making magnets – Induced magnetism



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- When the magnetic material (the nickel coin) is placed in the magnetic field of the permanent magnet (the horseshoe magnet), magnetism is induced in the coin and it becomes attached to the permanent magnet.
- The nickel coin is attracted to either the north or the south pole of the permanent magnet.
- When a coin is attracted to the south pole, a north pole is induced at the top of the coin and a south pole at the bottom. The opposite is true when a coin is attracted to the north pole.
- If the horseshoe magnet in was removed, the magnetic field would no longer be present, and the nickel coins would lose their magnetism (become unmagnetized or demagnetized)

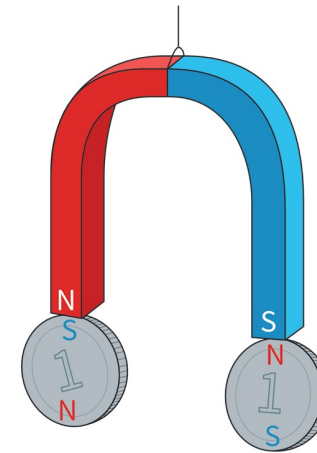


Figure 1. Induced magnetism in a nickel coin.

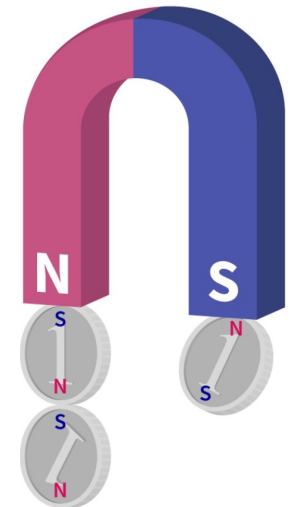


Figure 2. A horseshoe magnet with three coins attached

Hard and soft magnetic materials



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- Unlike nickel, some materials keep their magnetism after the external field is removed.
- A hard magnetic material, such as steel, is hard to magnetise but also hard to demagnetise/unmagnetise.
Steel is used to make permanent magnets in devices that require a constant magnetic field.
- A soft magnetic material, such as iron, is easy to magnetise but also easy to demagnetise/unmagnetise. Iron is a good material to use for temporary magnets. Iron is used in electronic door locks as it gains and loses its magnetism quickly.

Making magnets - Electromagnet



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- The figure at the right is known as an electromagnet. Any time a coil of wire carries current, a magnetic field is induced.
- Can you compare a permanent magnet and an electromagnet?

